

WebVMT: GIMI Video Metadata on the Web

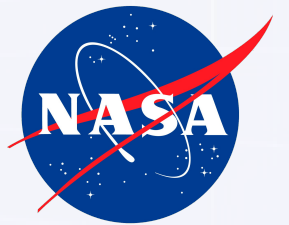
OGC Builder Days Code Sprint
Geovation, London, United Kingdom

Rob Smith | Away Team Software
13 May 2026

Meeting Host



Meeting Sponsor

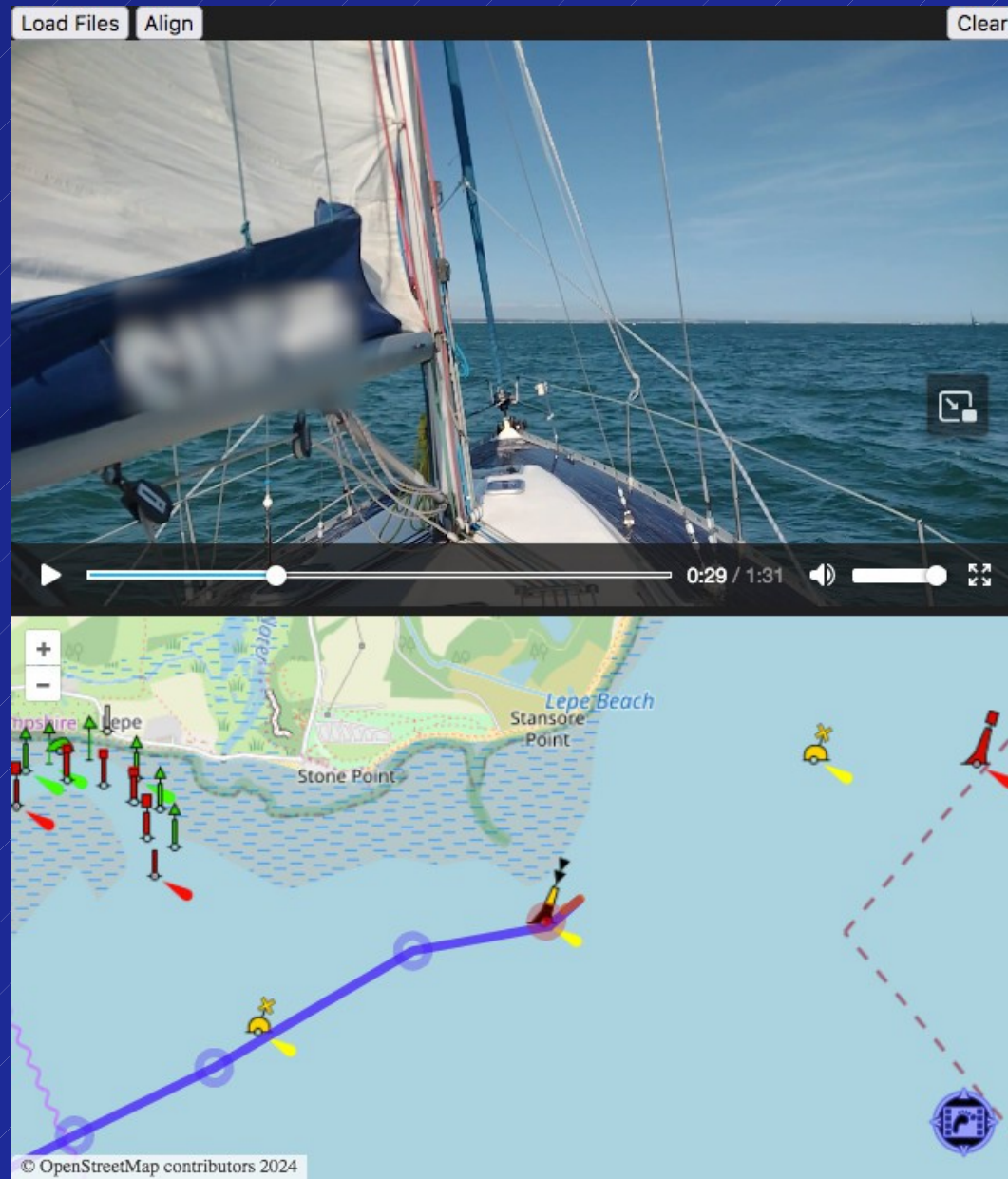


Agenda

- Video metadata basics
- How to extract metadata from a video file
- How to encode video metadata in GIMI
- How to share location metadata on the web
- How to access binary metadata on the web
- How to synchronize metadata in a web browser
- Dash cam collision monitoring demo
- Community resources

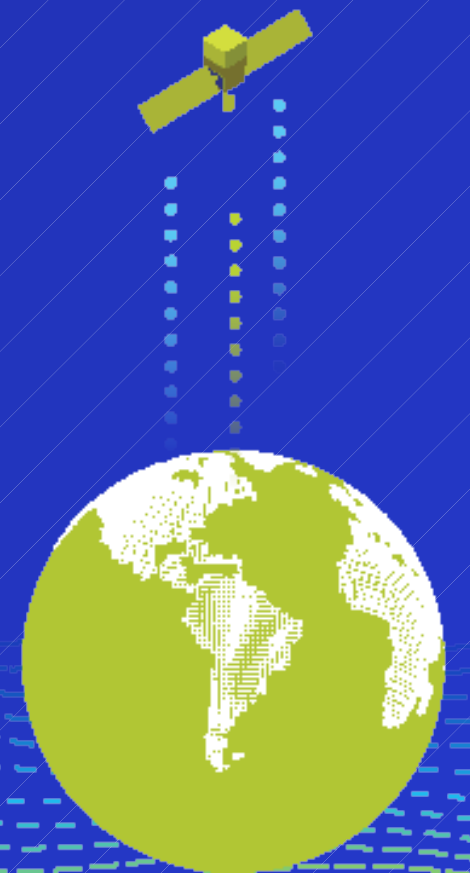


Dash cam collision monitoring



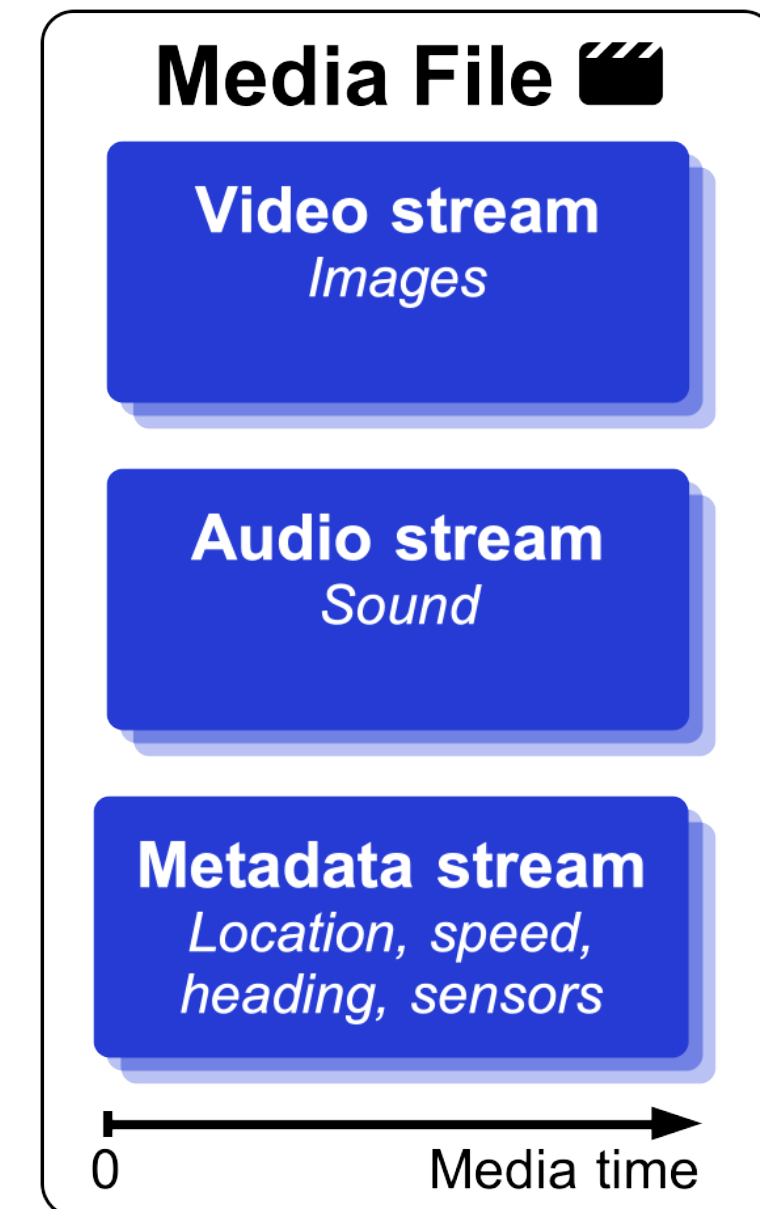
Video Metadata

GIMI Video Community Integration



Video Metadata Basics

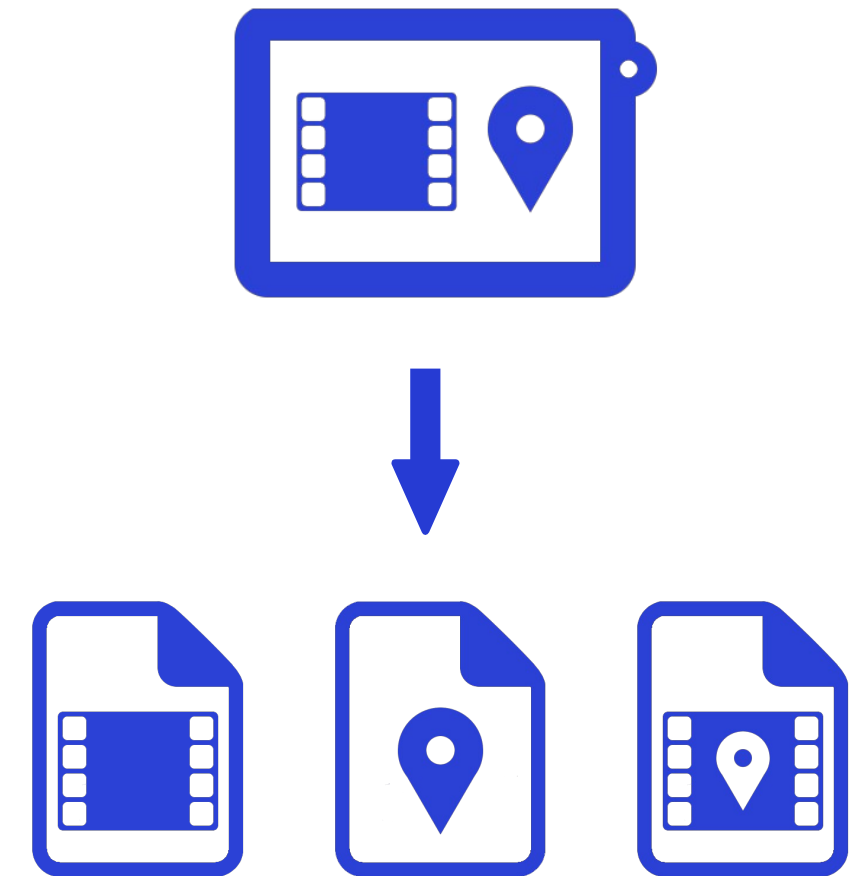
- Media file contents
 - Video
 - Audio
 - Metadata
- Streams synchronized by media file
 - Monotonically-increasing time
 - Immune to ambiguities such as leap seconds
 - Optionally referenced to date-time epoch



Media file content

Extracting Video Metadata

- Download data from camera
 - Proprietary or standard format
 - Sidecar file
 - Embedded file
- Extract metadata with community video tools
 - ExifTool: <https://exiftool.org/>
 - FFmpeg: <https://ffmpeg.org/>
 - libheif: <https://github.com/strukturag/libheif>
 - GStreamer: <https://gstreamer.freedesktop.org/>

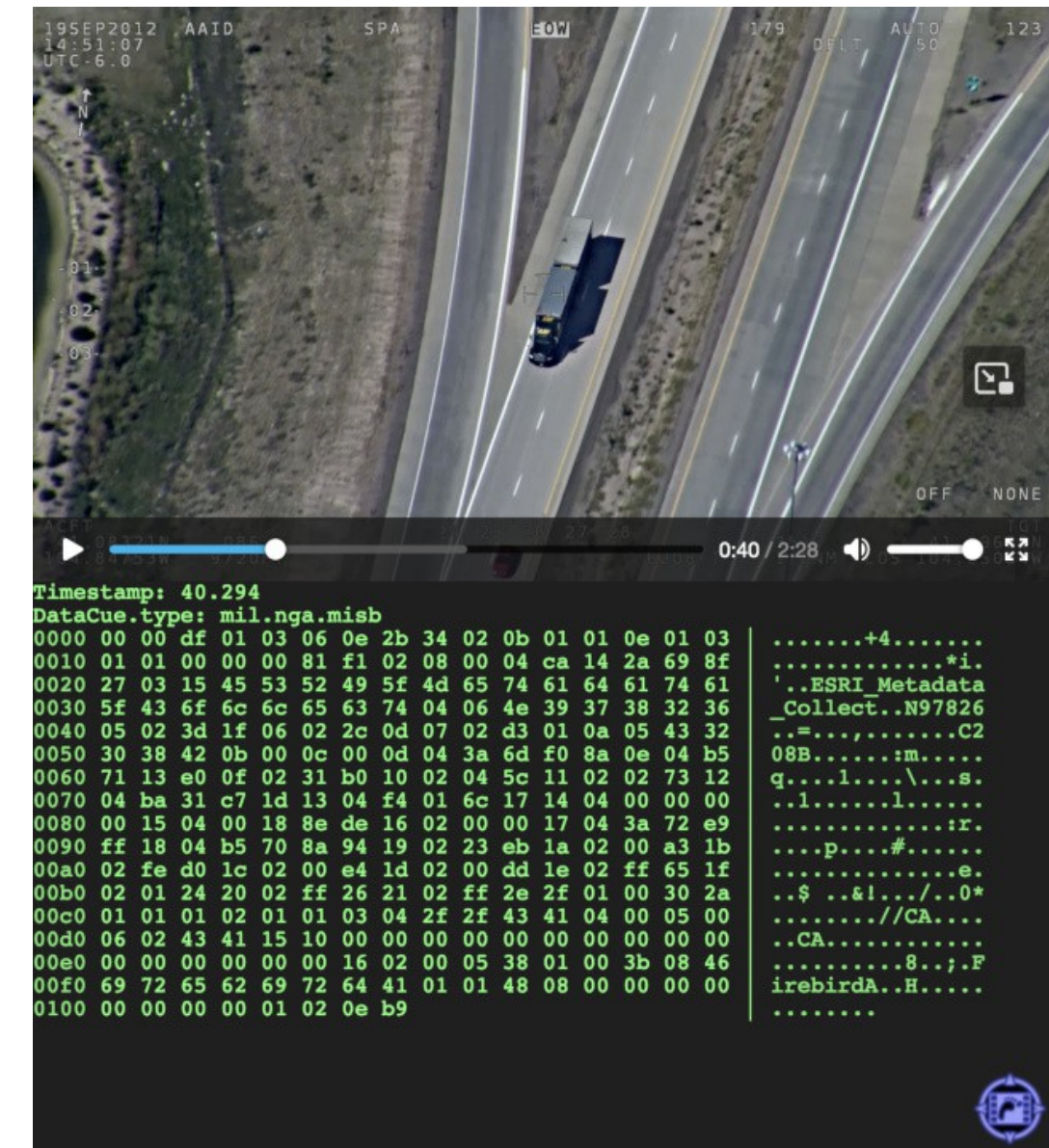


Encoding Video Metadata in GIMI

- OGC Testbed-21 GIMI
 - Integrated WebVMT support with libheif
 - Encode text or binary metadata with heif-enc
- Export video frames

```
ffmpeg -i input.mp4 frame%4d.png
```
- Encode metadata track

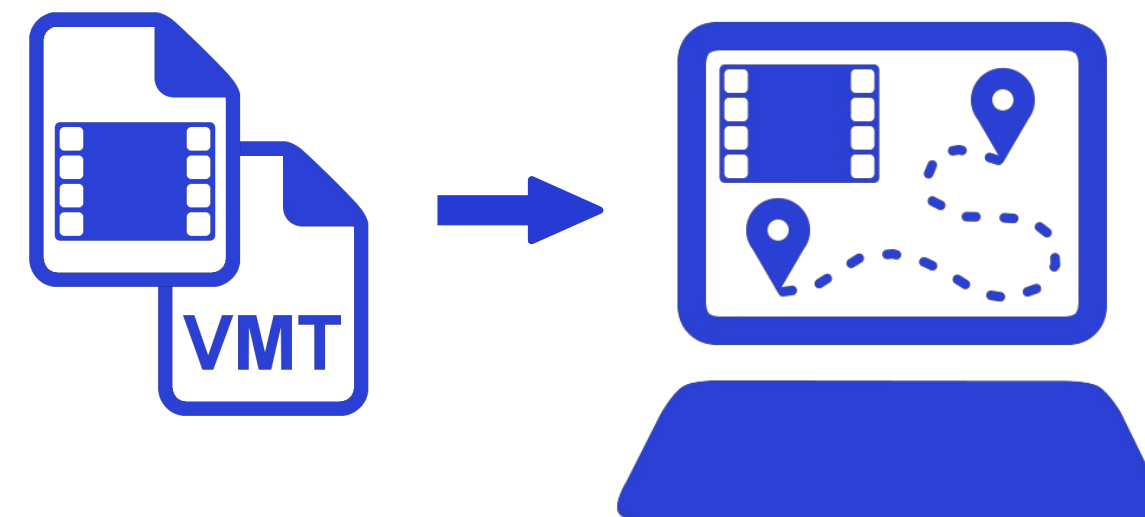
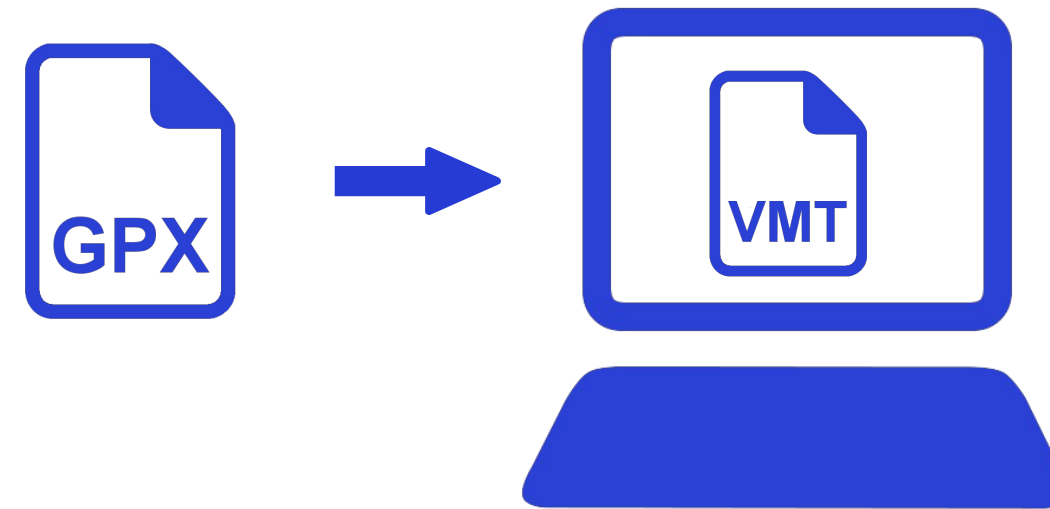
```
heif-enc -S frame0001.png -o output.heif \  
-V --vmt-metadata input.vmt
```



MISB metadata synchronized
with video on the web

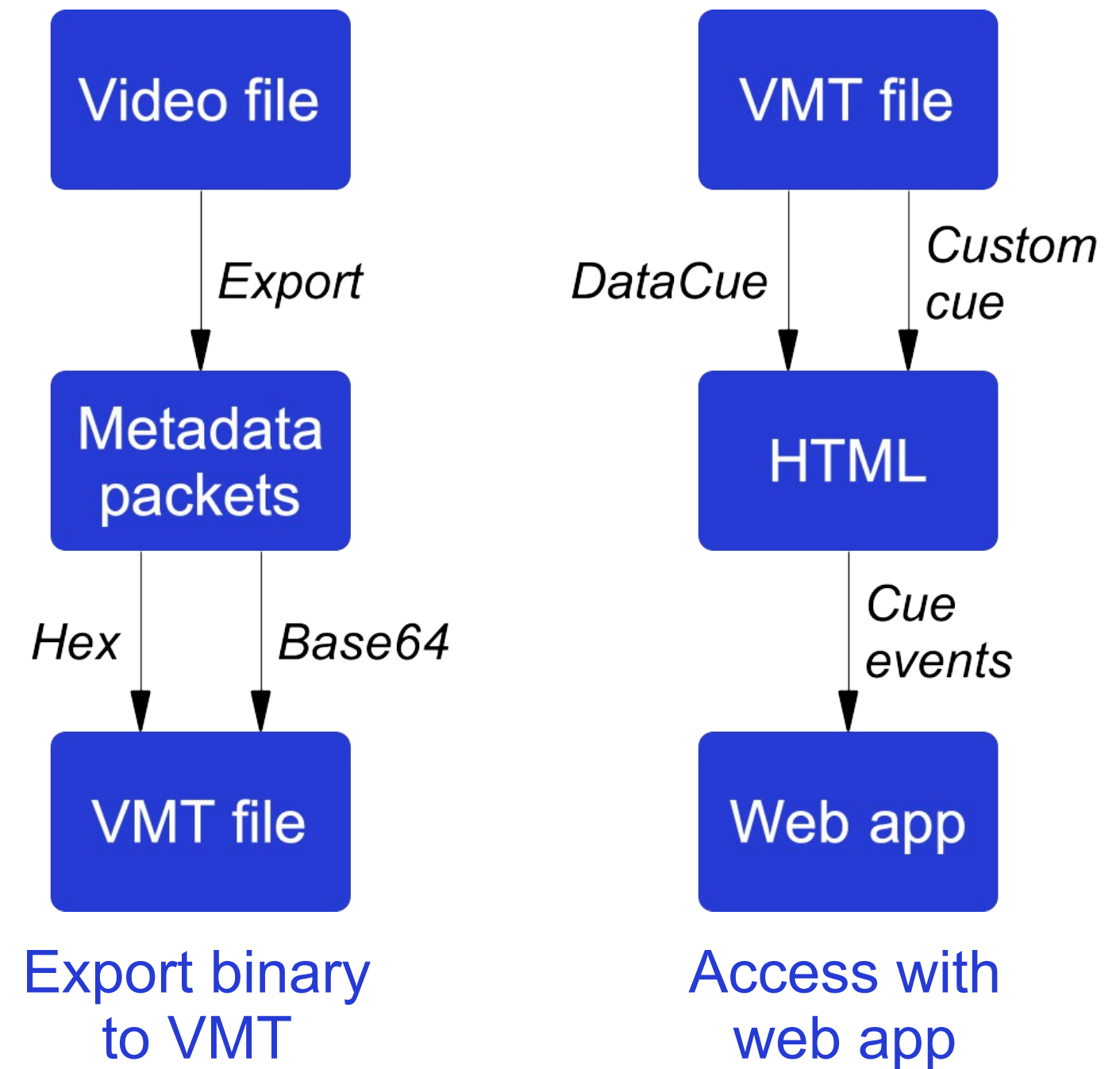
Sharing Location Metadata on the Web

- Export track to VMT
 - Export to GPX
 - Convert to WebVMT path
 - GPX Importer tool at webvmt.org
 - Drag & drop file to convert
- Display in a web page
 - Mobile demo at webvmt.org
 - Drag & drop files to display
 - Publish a custom web page



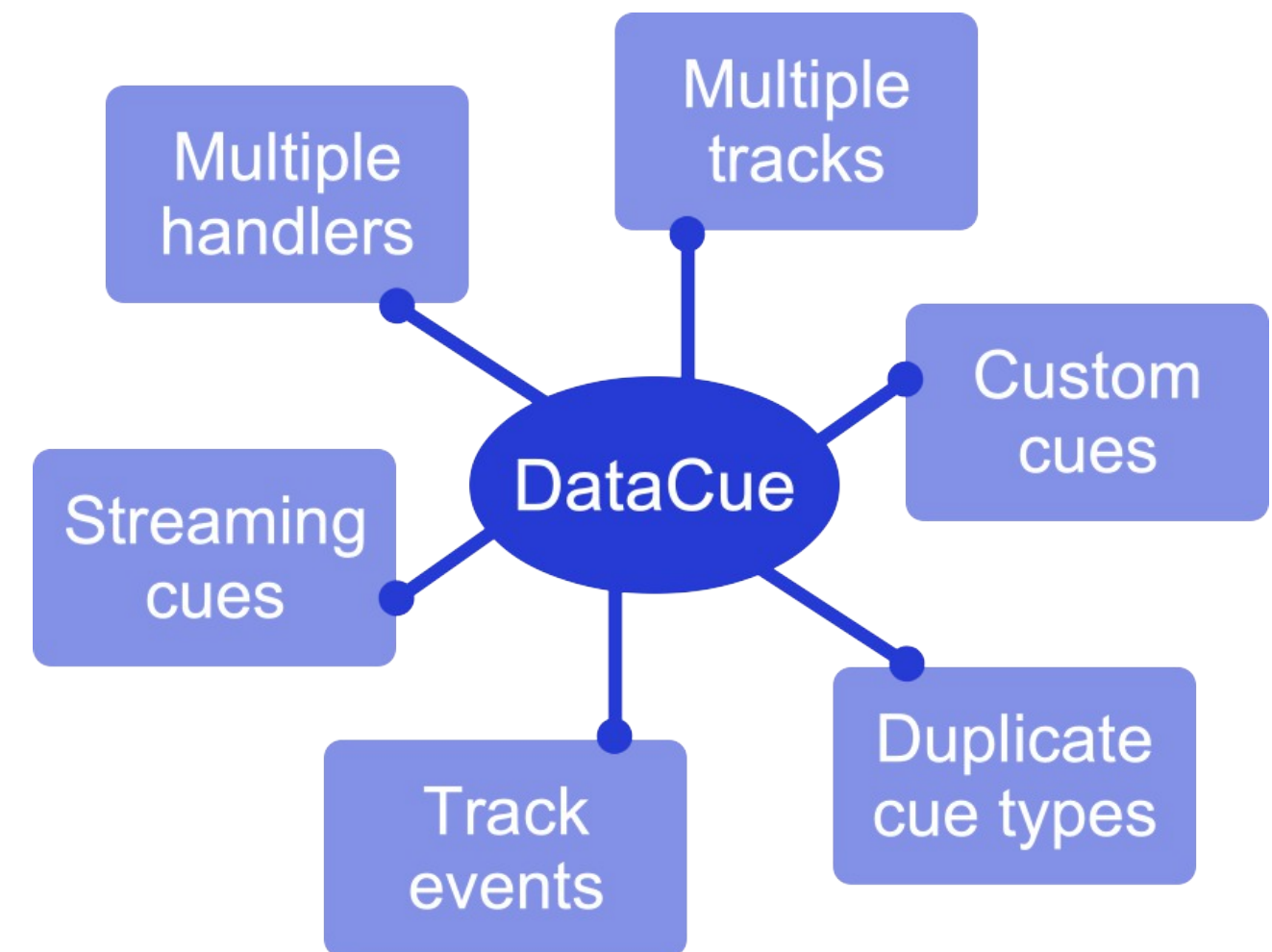
Accessing Binary Metadata on the Web

- Export binary to VMT
 - Encode with base64 or hex
 - Encapsulate with WebVMT sync command
 - <https://github.com/webvmt/community-group/issues/15>
- Access with web application
 - Synchronize with WebVMT
 - Access with HTML
 - DataCue or custom cue
 - Handle cue events
 - Filter by cue type



Handling Timed Metadata in a Web Browser

- HTML DataCue
 - **type**: identifier such as `mil.nga.misb`
 - **value**: metadata packet content
 - Integrated with WebVMT sync command
- DataCue implementation options
 - Live examples for developers
 - Comparison with WebVTT for timed text
 - <https://webvmt.github.io/vmt-sync-examples/>

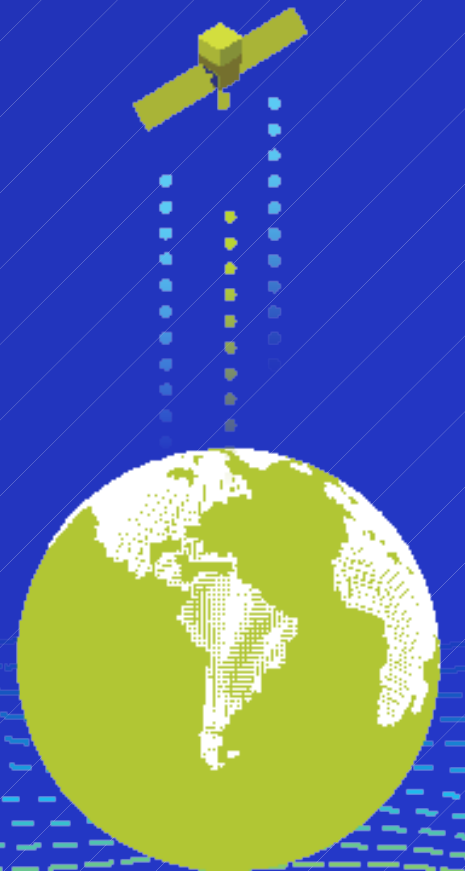


DataCue implementation options



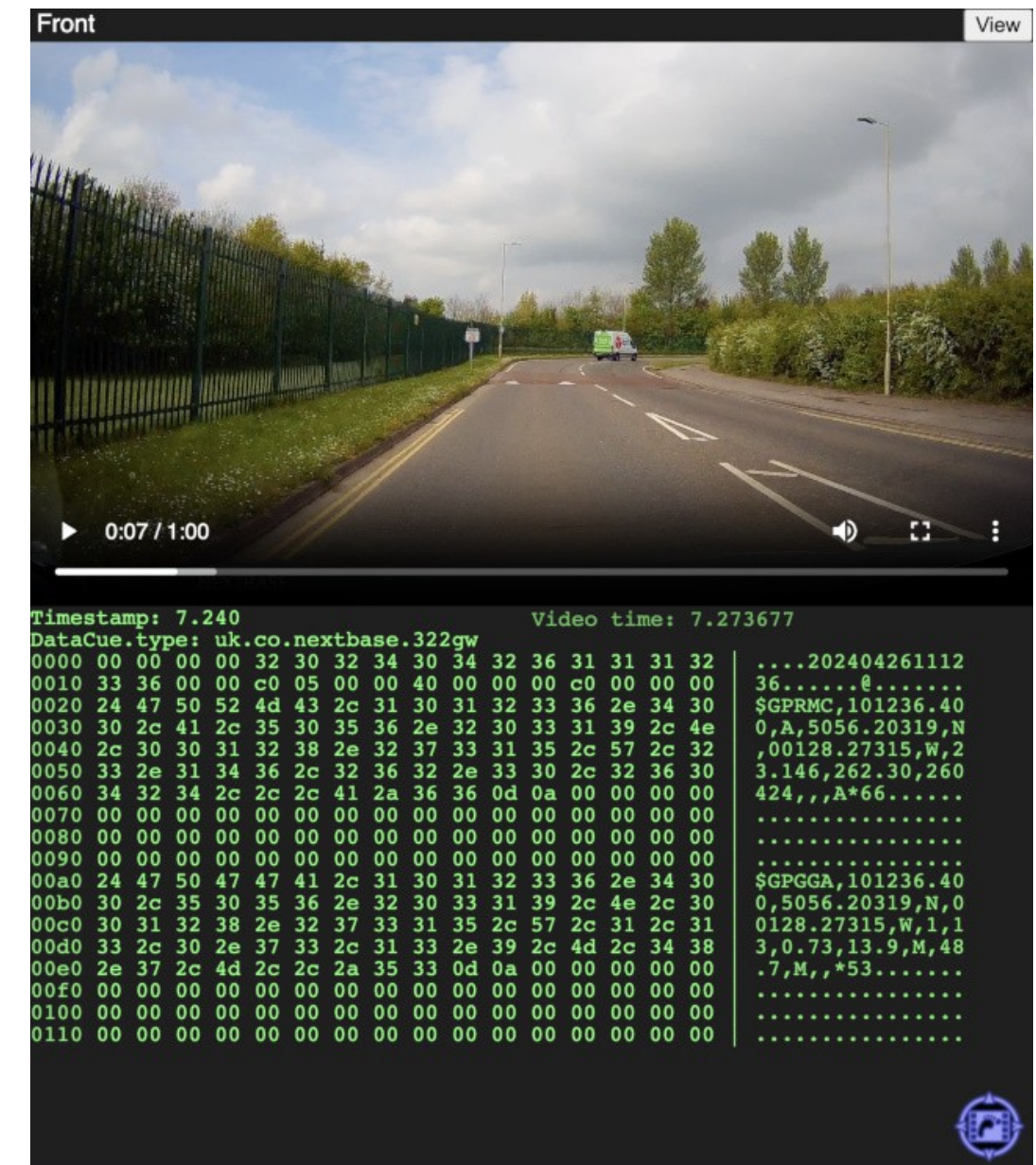
Collision Monitoring

Dash Cam Video Metadata



Dash Cam Collision Monitoring Use Case

- Demonstrate video metadata integration with web browsers for a commercial use case
- Identify admissible evidence for insurance claims and police investigations
- Protect employees and company vehicles
- Identify potholes for road maintenance from dash cam fleet
 - Identify events of interest from timed metadata
 - Associate with video and location
 - Analyze data using a web application in a browser



Binary metadata synchronized
with video in web browser

Dash Cam Collision Monitoring Use Case

- Corrected timing anomalies
- Separated location & accelerometer streams
- Combined & filtered metadata in web app
- Filter accelerometer data by axis & threshold or range
- Accurately identified speed bump locations & video from vertical accelerometer spikes (orange)
- Click location to show associated video frame



Accuracy of speed bump location
from accelerometer metadata

Conclusions

- Demonstrated how GIMI video metadata can be accessed in a web browser
 - Text & binary metadata formats
 - Agnostic of video encoding
 - Synchronized with video using WebVMT & DataCue
 - Integrated with HTML & CSS
 - Web application implemented on top of WebVMT
- Published details:
 - OGC Testbed-21 D206 Report [OGC 26-012]



Speed bump locations indicated by vertical accelerometer spikes

WebVMT Community Resources

- W3C Group Note
 - <https://www.w3.org/TR/webvmt/>
- Website
 - Demos, guides, tools & blog
 - <https://webvmt.org/>
- WebVMT Community Group
 - Ask questions, find answers & raise issues
 - <https://github.com/webvmt/community-group>
- Sync Command Examples
 - <https://webvmt.github.io/vmt-sync-examples>



THANK YOU

